

Simra Ali

Computer Engineering · Robotics, Autonomous Systems & Embedded Design

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#1 in MA · #11 in the world FTC 3565, out of 7,800+ teams (2026)	3× 1st place Hackathons & CAD competitions, 2024–25	GPA 4.18 · SAT 1560 790 Math · 770 EBRW	MIT BWSI · Harvard SSP Autonomous systems coursework
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EDUCATION

Westford Academy · Westford, MA	Class of 2027
GPA 4.18 · SAT 1560 (790 Math, 770 EBRW) · Academic High Honors, 2023–present	
Coursework: AP Calculus BC, AP Physics C, AP Chemistry, AP Computer Science A, AP Biology, AP U.S. History, AP Psychology, AP Microeconomics	
MIT Beaver Works Summer Institute · Lincoln Laboratory	Feb 2024 – Aug 2026
Autonomous Air Vehicle Racing	on-site · summer
Microelectronics & Hardware Development	online
Version Control: Git/GitHub & UNIX	online
Basics of ASICs	instructor-led · fall
Python Core	online
Harvard Summer School · Cambridge, MA	Summer 2025
Robotics, Autonomous Vehicles, Drones, and Artificial Intelligence — Grade: A	

ROBOTICS LEADERSHIP

FIRST Robotics Competition · Stormgears 5422 (45 members)	2023 – Present
Co-Captain · CAD / Mechanical Design Lead · Drive Team Coach · Former Electrical Lead	
- Taught myself SolidWorks after the team's experienced designer graduated, then helped lead a custom-robot effort that earned the FIRST Quality Award for design intent, robustness, and fabrication quality. - Designed competition mechanisms in SolidWorks — chassis, intake, hopper, shooter, electrical board, belly pan, and component holders — specified for serviceability between matches. - Ran summer and fall training for 20+ Electrical & CAD students and trained the student who succeeded me as Electrical Lead. - Managed inventory, schedule and task tracking (Slack, Trello). - Drive Team: Coach (2026 Season), Technician (2025 Season), Driver (2024 Clipper Clash). - Presented to sponsors including IBM, Triton Systems, Performance Motion Devices and SolidWorks.	
FIRST Tech Challenge · Ghost Robotics 3565 (10 members)	2023 – Present
Computer Vision Software Developer · Onshape CAD Designer	
2026 Massachusetts State Champion and Winning Alliance Captain. - Ross Division Winner at the FIRST World Championship: ranked #1 in Massachusetts and #11 in the world out of 7,800+ teams. - Work deliberately across vision software, CAD, strategy, media and outreach rather than limiting myself to one role.	

ENGINEERING PROGRAMS & RESEARCH

Autonomous Air Vehicle Racing · MIT Beaver Works Summer Institute, AeroAstro	Summer 2026
Four-week onsite program	
- Designed, programmed, repaired, tested and raced autonomous drones. - Studied computer vision, state estimation, control systems, navigation and flight dynamics. - Explored socially aware path planning for autonomous drones and connected with an MIT PhD researcher working in the field. - Honorable Mention, BWSI “You Belong in STEM” Essay Contest.	
Robotics, Autonomous Vehicles, Drones & AI · Harvard Summer School	Summer 2025
Grade: A	
- Studied sensors, motion planning, SLAM, reinforcement learning, autonomous systems and computer vision. - Co-led the “Solve It, Defend It!” innovation club and developed Flood Watch, a concept for predicting flood risk from drone and autonomous-vehicle data.	

Women in Technology Program · BAE Systems

Jan – May 2026

- Hands-on labs, workshops, tours and presentations spanning electrical, microwave/RF, software, mechanical, signal processing, optical, manufacturing and failure-analysis engineering.

Summer Immersion & Pathways Programs · Girls Who Code

Jun 2024 – Jul 2025

Cybersecurity, cryptography, data science, machine learning, game design and web development.

PROJECTS & COMPETITIONS

The Bat · GirlHacks at Microsoft NERD — 1st Place

Jun 2025

- Assistive navigation device for blind and low-vision users: ultrasonic sensors detect obstacles and a buzzer returns auditory feedback. Designed, built and demonstrated within the hackathon.

HireAble · NetScout Tech for Good — 1st Place

Mar 2025

- Web platform connecting people with disabilities to inclusive employers, integrating database and web technologies to make hiring more accessible.

VTXCADs CADathon · Phillips Exeter Academy Robotics — 1st Place

Dec 2024

Firefly · Girls Who Code game design project

Jul – Aug 2025

- Interactive game built with the p5play / p5.js JavaScript libraries, using TensorFlow.js hand-gesture detection as the input method.

SolidWorks Demonstration Robot · built for team sponsor

May – Sep 2024

- FRC-style robot with autonomous vision to locate tennis balls, an intake to collect them, and a shooter to launch them at a target.

OUTREACH, MENTORSHIP & LEADERSHIP

Outreach Coordinator · Middlesex County Robotics Club, Boxboro, MA

Aug 2023 – Present

- Organize and lead Girl Scout STEM Fest, FIRST LEGO League trainings and scrimmages, the Apple Blossom Parade, and elementary and middle school STEM events; emphasis on getting girls into technical roles and keeping them there.
- Led the upgrade of the SaturnReach outreach robot.

Co-Founder & Co-Lead · Women in Computer Science, Westford Academy

Sep 2024 – Present

- Teach Python, support member projects, run hackathon preparation, and share STEM opportunities that are easy to miss if nobody tells you they exist.

Manager · Westford Academy Broadcasting Club

Sep 2024 – Present

- Coordinate club operations while working the shows as anchor and production crew: teleprompter, sound board, video switching, scripts and school-wide broadcasts.

Cashier · Demoulas (Market Basket), Westford, MA

Jul 2023 – Dec 2024

HONORS & AWARDS

FIRST Leadership Award (Dean's List) Semi-Finalist	2026	Academic High Honors, Westford Academy	2023–26
BWSI "You Belong in STEM" Essay — Honorable Mention	2026	AP Scholar Award	2026
GirlHacks at Microsoft NERD — 1st Place	2025	FRC Quality, Team Spirit & Gracious Professionalism	2025–26
NetScout Tech for Good — 1st Place	2025	FTC Control, Inspire, Innovate & Connect	2025–26
VTXCADs CADathon — 1st Place	2024	Diversity, Equity & Inclusion Award, Westford	2023
Toastmasters Youth Leadership Award & Medal	2021	Honored Citizen Award, Westford Public Schools	2023

TECHNICAL SKILLS

CAD / Mechanical SolidWorks, Onshape, 3D printing, Autodesk Tinkercad

Hardware Arduino, wiring, soldering, crimping, electrical-board layout, sensors, systematic troubleshooting

Software Python, Java, JavaScript, HTML/CSS, SQL, Git/GitHub, WPILib, TensorFlow, NumPy, p5.js

Robotics / AI Computer vision, SLAM, motion planning, reinforcement learning, autonomous systems, state estimation